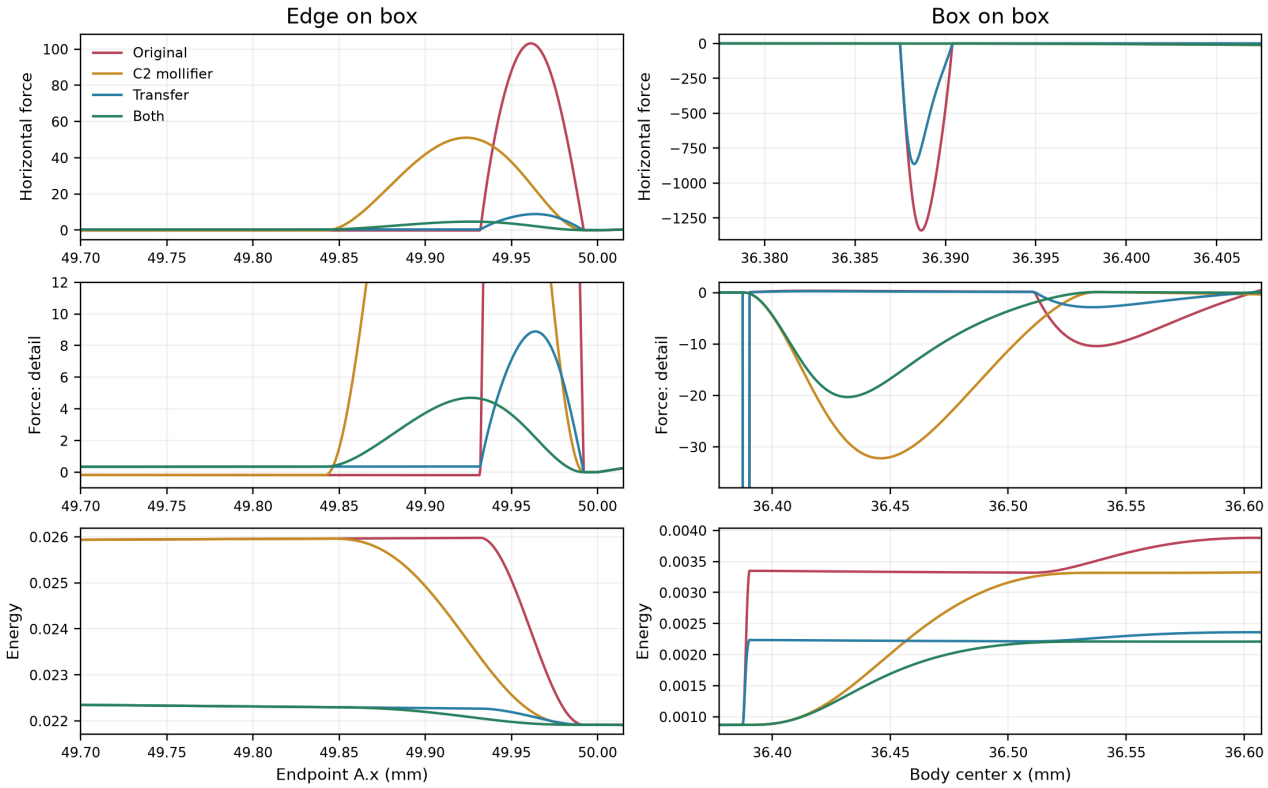
ESP force-spike experiments

Result. Combining local quadrature transfer with the proposed wider C2 mollifier reduces the sampled horizontal-force peak by **95.44% for edge on box** and **98.48% for box on box**. Residual peaks remain.

Experiment. Both original issue scenes retain their geometry and contact parameters. The proposed mollifier replaces each endpoint factor with H(c/ell), where H(t) = 10t^3 - 15t^4 + 6t^5 for 0 < t < 1 (clamped to 0/1 outside), c is endpoint clearance and **ell = 150 micrometers**. Distance support is unchanged. All four variants differentiate their complete energy.

Variant	Edge peak |Fx|	Reduction	Box peak |Fx|	Reduction
Original ESP	103.13	-	1338.85	-
New C2 mollifier only	51.05	50.49%	32.22	97.59%
Local transfer only	8.89	91.38%	862.79	35.56%
Transfer + C2 mollifier	4.70	95.44%	20.32	98.48%



Top: full spike scale. Middle: reduced-force detail (larger curves clipped). Bottom: energy. The box detail and energy panels use a wider x range to show the broadened transition. Edge forces act on endpoint A; box forces are summed over the moving box. Values use each original reproduction's scaling.

Sampling and checks. Edge: 49.700-50.015 mm, 0.5 micrometer base spacing. Box: 35.58749-37.18749 mm, 1 micrometer base spacing plus the original 0.025 micrometer spike grid. Each variant's maximum is locally refined; reported peaks remain sampled maxima. Energy reconstruction matches compiled old/transfer evaluation throughout. Peak forces agree at FD steps 0.2 and 0.1 nm; old/transfer forces also match compiled gradients at all four peak locations.

Interpretation. The wider mollifier contributes most of the box-on-box improvement; transfer alone leaves a large spike. Combining both performs best in each scene. The C2 variants remain diagnostic prototypes and change the energy and transition width. Refined sampling slightly updates the previous peak values.

Evidence: results/implementation/four-variants/{edge,box}.csv and {edge,box}-summary.json. Rerun: .venv-arm64/bin/python diagnostics/compare_four_variants.py. All four variants were evaluated anew for this report.